

Web-Based Water-Level Monitor and Pump Controller



M. KATHIRESAN, A. ROBSON BENJAMIN
AND C. VIJAYAN

Several articles about water-level controllers for overhead tanks have been published in earlier issues of EFY magazine. But the project we are going to share today is new and unique. So presented here is a Web-based water-level monitor and motor-pump controller using ATmega128 microcontroller, Ethernet controller ENC28J60 and TCP/IP protocol. The design involves the use of

port-forwarding facility of broadband modem or router. By using this facility, one can port forward a local IP address to a public IP address with which controlling appliances from any part of the world through the Web becomes possible. The authors' prototype is shown in Fig. 1. Five water-level sensors are used to monitor the water levels in the overhead tank via Web browser.

Circuit and working

The circuit diagram of the Web-based water-level monitor and pump controller is shown in Fig. 2. The circuit is built around microcontroller ATmega128 (IC1), Ethernet controller ENC28J60 (IC2), a sensor circuit and 16×2 LCD display.

ATmega128. ATmega128 microcontroller of AVR family is chosen for

the design as it is inexpensive and has sufficient memory to carry out the necessary tasks. ATmega128 has 128k Flash memory, 4k EEPROM and 4k SRAM. It is available in 64-pin TQFP package and has SPI hardware. Even though it is not difficult to solder 64-pin TQFP package, off-the-shelf available ATmega128 module (Fig. 3) is used to speed up the design process.

The code is written in C and compiled using AVR-GCC to fit into the memory size of ATmega128. The HTML code is put into program memory of ATmega128; therefore the use of external EEPROM is avoided.

ENC28J60 module. ENC28J60 Ethernet controller from Microchip is a low-cost and easy-to-handle IC since it comes with a 28-pin DIP package having a serial peripheral interface (SPI) bus. A number of Ethernet controllers are available in the market but Microchip Ethernet controller is easy to integrate with any other controller. It operates at 3.3V power supply and is readily available in the market. Details of the driver and its functions are provided in the Microchip TCP/IP stack section of <http://www.microchip.com>

website. The driver functions have been implemented in this design after careful study.

SPI communication. ATmega128 microcontroller is chosen as master and ENC28J60 as slave for the SPI communication. FOSC/2 clock mode is chosen by properly setting SPSR register in the microcontroller. Since ENC28J60 op-

PARTS LIST

Semiconductors:

IC1	- ATmega128 microcontroller
IC2	- ENC28J60 Ethernet controller
IC3	- 74HC125 level shifter
IC4	- 7805, 5V regulator
T1-T5	- BC547 npn transistor
D1	- 1N4001 rectifier diode
ZD1	- 3.3V, 0.5W zener diode
BR1	- 1A bridge rectifier module
LED1-LED5	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):

R1, R4	- 680-ohm
R2	- 27-ohm, 0.5W
R3, R6-R9, R17-R25	- 220-ohm
R5	- 10-kilo-ohm
R10	- 2.7-kilo-ohm
R11, R12	- 470-ohm
R13-R16	- 49.9-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 470µF, 25V electrolytic
C2, C3, C6	- 0.1µF ceramic disc
C4, C5	- 10µF, 16V electrolytic
C7, C8	- 18pF ceramic disc
C9	- 0.01µF ceramic disc
C10-C11	- 22pF ceramic disc

Miscellaneous:

X1	- 230V AC primary to 9V, 500mA secondary transformer
X _{TAL1}	- 16MHz crystal oscillator
X _{TAL2}	- 25MHz crystal oscillator
Motor	- 230V AC water pump motor
RL1	- 5V, 1/CO relay
LC1	- 16×2 LCD module
S1	- Tactile switch
CON1	- RJ45 Ethernet connector
CON2	- 6-pin header/connector
CON3, CON4	- 2-pin, 5mm terminal connector

TABLE I
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	3.3V
TP3	Low when S1 is pressed

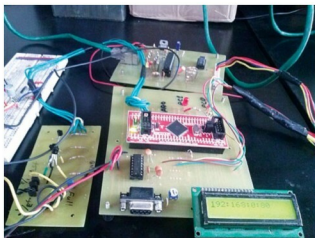


Fig. 1: Authors' prototype of Web-based water-level monitor and pump controller